

# LOVELY PROFESSIONAL UNIVERSITY

## Sample Question Set - Unit 2

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO2

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 2

Unit Title: Partition-Based Clustering

Unit Mapped RBT: L4 (Analyze)

Total Questions: 40

Suggested Duration: 60 Minutes

### Answer Grid (to be filled by the student)

| Q. No. | Answer | Q. No. | Answer | Q. No. | Answer | Q. No. | Answer |
|--------|--------|--------|--------|--------|--------|--------|--------|
| Q1     |        | Q11    |        | Q21    |        | Q31    |        |
| Q2     |        | Q12    |        | Q22    |        | Q32    |        |
| Q3     |        | Q13    |        | Q23    |        | Q33    |        |
| Q4     |        | Q14    |        | Q24    |        | Q34    |        |
| Q5     |        | Q15    |        | Q25    |        | Q35    |        |
| Q6     |        | Q16    |        | Q26    |        | Q36    |        |
| Q7     |        | Q17    |        | Q27    |        | Q37    |        |
| Q8     |        | Q18    |        | Q28    |        | Q38    |        |
| Q9     |        | Q19    |        | Q29    |        | Q39    |        |
| Q10    |        | Q20    |        | Q30    |        | Q40    |        |

### Multiple Choice Questions (40 x 1 = 40 Marks)

**Q1. State what the letters WCSS stand for in the reporting of a k-Means result.**

- A) Within-Cluster Sum of Squares
- B) Weighted Cluster Similarity Score
- C) Whole Cluster Separation Score
- D) Within-Cluster Silhouette Sum

**Q2. Identify the shape of the region of space that a single k-Means cluster occupies.**

- A) A ring around one centre
- B) A convex region around one centre
- C) A curved band of any shape
- D) A set of disconnected patches

**Q3. Name the two initialisation strategies given on the syllabus for choosing the starting centres of k-Means.**

- A) Random initialisation and the elbow method
- B) The elbow method and the silhouette method
- C) Random initialisation and k-Means++
- D) Standardisation and min-max scaling

**Q4. Identify what the abbreviation PAM stands for in the name of the k-Medoids algorithm.**

- A) Partitioning And Merging
- B) Proximity Assignment Method
- C) Partition Average Minimisation

D) Partitioning Around Medoids

**Q5. Identify the quantity that k-Medoids sets out to minimise.**

- A) The sum of the squared distances to the mean
- B) The sum of the distances to the medoid
- C) The number of records in the largest cluster
- D) The distance between the two nearest medoids

**Q6. State the inertia of a clustering in which k equals the number of distinct records.**

- A) One
- B) The number of records
- C) It cannot be computed
- D) Zero

**Q7. Explain why a k-Means centroid need not coincide with any record of the dataset.**

- A) It is an average, not a chosen record
- B) It is always placed at the origin
- C) It is chosen before the data is read
- D) It is the record furthest from the rest

**Q8. Distinguish the k-Means cost from the k-Medoids cost by the power of distance each one uses.**

- A) k-Medoids squares the distances, k-Means does not
- B) Both of the two square the distances used
- C) k-Means squares the distances, k-Medoids does not

D) Neither of the two squares the distances used

**Q9. Interpret a negative silhouette coefficient reported for a record.**

- A) It sits exactly at the centre of its cluster
- B) It is equidistant from every cluster centre
- C) It sits nearer to a neighbouring cluster than its own
- D) It has been excluded from every cluster

**Q10. Describe what min-max scaling produces from a numeric column.**

- A) Values lying between 0 and 1
- B) Values with mean 0 and variance 1
- C) Values lying between -1 and +1
- D) Values converted into whole numbers

**Q11. Predict what must happen if k is set larger than the number of distinct records in the table.**

- A) The algorithm splits records into halves
- B) The extra centres are placed at the origin
- C) Every cluster receives the same record
- D) At least one cluster must be left empty

**Q12. Recognize the situation described: during an iteration one of the centres ends up with no points assigned to it at all.**

- A) A converged cluster, which is left alone
- B) An empty cluster, which must be reinitialised
- C) A noise cluster, which is reported as noise
- D) A merged cluster, which absorbs its neighbour

**Q13. Explain why an inertia value from a scaled table cannot be compared with one from the same table unscaled.**

- A) Inertia is undefined once a table is scaled
- B) Inertia always falls to zero after scaling
- C) Inertia is computed from the labels, not features
- D) Inertia is expressed in the units of the features

**Q14. Distinguish the within-cluster sum of squares from the between-cluster sum of squares.**

- A) Within measures separation, between measures compactness
- B) Both of the two measure the compactness of clusters
- C) Within measures compactness, between measures separation
- D) Both of the two measure the separation of clusters

**Q15. Explain why MiniBatch k-Means normally finishes with slightly higher inertia than full k-Means.**

- A) It uses a different distance measure entirely

- B) Its updates are computed from a noisy sample
- C) It always stops after a single iteration
- D) It refuses to move a centre once it is set

**Q16. Interpret a Davies-Bouldin Index that is very close to zero.**

- A) The clusters are compact and well separated
- B) The clusters overlap almost completely
- C) The clustering has failed to converge
- D) The number of clusters is far too large

**Q17. Predict the effect on k-Means of scaling only one of the two numeric columns.**

- A) Both columns then contribute equally
- B) The column left unscaled continues to dominate
- C) The scaled column takes over the result
- D) The algorithm can no longer be run

**Q18. Describe what k-Means assumes about the relative sizes and densities of the groups it looks for.**

- A) That they are of broadly similar size and spread
- B) That one group is far larger than the rest
- C) That every group has exactly the same shape
- D) That the groups may overlap without limit

**Q19. Solve the following: a cluster holds the two-dimensional points (1, 2), (3, 4) and (5, 6). Determine its centroid.**

- A) (2, 3)
- B) (4, 5)
- C) (9, 12)
- D) (3, 4)

**Q20. Compute the inertia of a one-dimensional cluster holding 4, 6 and 8 whose centre lies at 6.**

- A) 4
- B) 6
- C) 8
- D) 12

**Q21. Apply the silhouette definition to a record whose mean distance to its own cluster is 3 and to the nearest other cluster is 5.**

- A) 0.4
- B) 0.6
- C) 1.7
- D) 0.2

**Q22. Use the k-Medoids cost to determine the medoid of the one-dimensional cluster holding 2, 3 and 10.**

- A) 2
- B) 10
- C) 3
- D) 5

**Q23. Solve the following: a feature has mean 100 and standard deviation 20. Determine the standardised value of an observation equal to 70.**

- A) -0.5
- B) -1.5
- C) 1.5
- D) -3.0

**Q24. Compute the number of point-to-centre distance calculations made by one assignment step on 2,000 records with k equal to 5.**

- A) 2,005
- B) 400
- C) 25,000
- D) 10,000

**Q25. Use min-max scaling to determine the scaled value of 30 in a column whose minimum is 10 and maximum is 50.**

- A) 0.3
- B) 0.6
- C) 0.5
- D) 0.2

**Q26. Apply the elbow method to the following inertia readings and determine the value of k to report: k = 1 gives 1200, k = 2 gives 500, k = 3 gives 200, k = 4 gives 180, k = 5 gives 172.**

- A) 2
- B) 4
- C) 5
- D) 3

**Q27. Examine the following and select the correct explanation: a k-Means run on a table of 500 records reports an inertia of exactly zero.**

- A) Every record has been made a cluster of its own
- B) The features were standardised beforehand
- C) The algorithm stopped before converging
- D) The distance measure was chosen wrongly

**Q28. Solve the following: a two-dimensional cluster holds the points (0, 0) and (4, 6). Determine its centroid.**

- A) (4, 6)
- B) (2, 3)
- C) (0, 0)

- D) (2, 6)

**Q29. Apply standardisation to a numeric column and determine the mean and standard deviation of the result.**

- A) Mean 1 and standard deviation 0
- B) Mean 0 and standard deviation 0
- C) Mean 0.5 and standard deviation 0.5
- D) Mean 0 and standard deviation 1

**Q30. Apply the k-Means++ rule to two candidate points lying at distances 3 and 9 from the centre already chosen, and determine the probability that the point at distance 9 is selected next.**

- A) 0.75
- B) 0.9
- C) 0.5
- D) 0.6

**Q31. Examine the following and select the correct explanation: after the analyst standardises the table, the k-Means clusters change completely.**

- A) The number of clusters was silently increased
- B) Standardisation removes records from the table
- C) Every feature now contributes comparably to the distance
- D) The algorithm switched to a different objective

**Q32. Compare one iteration of full k-Means with one iteration of MiniBatch k-Means on a dataset of one million records.**

- A) MiniBatch touches only the records of its batch
- B) MiniBatch touches every record twice over
- C) Both of the two touch every record once
- D) MiniBatch touches more records than full k-Means

**Q33. Analyze why the k-Means objective is certain to fall at every step yet is not certain to reach its lowest possible value.**

- A) Each step reduces inertia but only a local minimum is reached
- B) Each step raises inertia until convergence is forced
- C) The objective is convex, so the minimum is always found
- D) The objective is fixed once the centres are initialised

**Q34. Differentiate the effect of a single extreme outlier on a k-Means centroid from its effect on a k-Medoids medoid.**

- A) The medoid is pulled towards it, the centroid is not

- B) Both of the two are pulled towards it equally
- C) Neither of the two is affected by an outlier
- D) The centroid is pulled towards it, the medoid is not

- C) Both of the two show that the record is borderline
- D) Neither of the two carries information about boundaries

**Q35. Infer what it indicates when the silhouette score falls steadily as k is raised from 2 upwards.**

- A) The data holds a very large number of groups
- B) The data holds little genuine group structure
- C) The features have all been standardised twice
- D) The algorithm has failed to converge at any k

----- End of Sample Question Set -----

**Q36. Distinguish what inertia tells you about a partition from what the silhouette coefficient tells you.**

- A) Inertia reports separation only, silhouette reports compactness
- B) Both of the two report compactness and separation
- C) Inertia reports compactness only, silhouette reports separation too
- D) Both of the two report compactness and nothing else

**Q37. Analyze the consequence of clustering a table in which one informative column has accidentally been included twice.**

- A) The duplicate column is detected and removed
- B) The distances between records become zero
- C) That quantity counts twice in every distance
- D) The number of clusters is automatically doubled

**Q38. Infer why the elbow method and the silhouette score can point to different values of k on the same data.**

- A) One of the two is always computed incorrectly
- B) They measure different properties of a partition
- C) They can only differ when the data is unscaled
- D) The silhouette is undefined for small values of k

**Q39. Relate the number of restarts used by a k-Means run to how much the reported partition can be trusted.**

- A) More restarts guarantee the global minimum is found
- B) More restarts reduce the number of clusters needed
- C) More restarts remove the need to scale the features
- D) More restarts make an unlucky single start less likely to survive

**Q40. Differentiate a hard assignment from a soft assignment by what each conveys about a record lying on a boundary.**

- A) The soft assignment shows the record is borderline, the hard one hides it
- B) The hard assignment shows the record is borderline